



亞洲大學
ASIA UNIVERSITY

Midterm Project Report

Advanced Computer Programming

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Chapter 1 Introduction

1.1 Github

- 1) **Personal Github Account:** <https://github.com/Jeffy-jpg>
- 2) **Group Project Repository:** <https://github.com/wangyalal/Group1>

1.2 Overview

In my program, I used Scrapy to scrape the data from Git Hub. Along with that, I imported datetime, and item adapter to adjust the format of the date. My program extracts the URL, About, Languages, Commits, time of the Last Update and the Name of the repository on the page <https://github.com/trungvdhp> . If the Repository did not have any information under the 'About' section, then it was replaced with the 'Name' of the repository. If there was no information under 'languages' and 'commits', then it was set to 'none'.

The scrapped information is then printed out into an XML file.

Chapter 2 Implementation

2.1 Class 1:

Description of Functional Class:

Within this project, two types of classes were used: Functional Class and a Data class. This sections (2.1) talks about the Functional and main Class of the of program. It uses the import statements scrapy, and github_scraper.items. Scrapy is used to write the code to scrape the the website and github_scraper.items is used to save the each field to an item for efficiency.

2.1.1 Method 1:

The parse Method:

This is used to pull the information from the page

<https://github.com/trungvdhp?tab=repositories> and save the section that holds the repositories to the variable 'repos' .

A loop is then created to get the link to each repository on the page and to extract the data for the last update. The spider then uses that link and both calls on the second method while also passing the information from the 'last_update' to the second method using cb_kwargs.

2.1.2 Method 2:

The step_in_repo method:

This is used to pull the information from within each repository, format the necessary information and display it. After the function is called, the scrapy item GithubItem is created to store the scraped information. After the information is stored, it's then displayed using 'yield'

2.2 Class 2

The purpose of this class is to save the data from the Functional class to an item so it's easier to print out the data using the single item rather than typing out the fields every time you want to output the data.

2.2.1 Fields

A scrapy item named `repo_info` with the following fields:

- Commits: String
- Languages: String
- Name: String
- About: String
- URL: String
- Last_Update: String

2.2.2 Methods

No method since it's a dataclass

2.2.3 Functions

No function since it's a dataclass

2.3 Class 3: GithubScraperPipelines

This class is a functional class responsible for cleaning the data collected from the last update and formats it to a human readable format.

2.3.1 Method 1:

In the class `GithubScraper`, the information within the scrapy item `repo_info` is passed into the method.

Using an adapter, the information is transferred into the class and the time saved under "Last_updated" is collected, cleaned and formatted to show the year, month, day and time in that specific order. After the information is cleaned, it's then returned to the scrapy item and furthermore, returned from the class.

Chapter 3 Results

3.1 Result 1

```
?xml version="1.0" encoding="utf-8"?>
```

```
<items>
```

```
<item><Commits>59
```

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Chapter 4 Conclusions

The main point of this program is to test and practice our ability to scrape information from a github page. Information such as the Url, About, Languages, Number of Commits, Last Update Time and the Name of the repositories were scraped and displayed as an XML document. During the writing of this code, I was able to learn not only the frame work of scrapy but also how it differs from beautifulsoup and some of the limitations that presented itself while trying to write the code. Using what I learnt from this project, I would work with my group to add more features including logging into a website on the user's behalf and automatically scraping and displaying the information without having to refer to the terminal.